

## 15. Performance Testing

|                      |                                                     |
|----------------------|-----------------------------------------------------|
| <b>Date</b>          | 1 July 2026                                         |
| <b>Team ID</b>       | 6a3a555ae5895b2c9759029d                            |
| <b>Project Name</b>  | EduGenie – Google Gemini Powered Learning Assistant |
| <b>Maximum Marks</b> | 5 Marks                                             |

Performance testing was carried out locally against the running FastAPI application using its TestClient, issuing 15 sequential requests per endpoint and measuring wall-clock response time. Because no GEMINI\_API\_KEY was configured for this test run, all requests were served from EduGenie's built-in demo/mock mode — this isolates and measures the overhead of the FastAPI routing, Pydantic validation and JSON-file logging layers, independent of live Gemini network latency.

### Step 1: Testing Overview

| Testing Tool Used   | Python (time.perf_counter) via FastAPI TestClient, 15 requests per endpoint       |
|---------------------|-----------------------------------------------------------------------------------|
| Type of Testing     | Baseline response-time measurement (demo/mock mode)                               |
| Target Module / API | /qa, /api/explain, /api/quiz, /api/summary, /learn/recommendations, / (home page) |
| Test Environment    | Local container, single process, no live Gemini network calls                     |
| Test Date           | 10 July 2026                                                                      |

### Step 2 & 3: Test Scenarios and Results (15 requests each, demo/mock mode)

| Endpoint                   | Avg Response Time | Min     | Max      | Status                                                     |
|----------------------------|-------------------|---------|----------|------------------------------------------------------------|
| GET /qa                    | 5.02 ms           | 3.79 ms | 16.60 ms | Pass                                                       |
| POST /api/explain          | 4.36 ms           | 4.03 ms | 5.16 ms  | Pass                                                       |
| POST /api/quiz             | 8.90 ms           | 7.51 ms | 13.81 ms | Pass                                                       |
| POST /api/summary          | 5.68 ms           | 5.14 ms | 7.13 ms  | Pass                                                       |
| GET /learn/recommendations | 5.72 ms           | 5.35 ms | 6.37 ms  | Pass                                                       |
| GET / (home page)          | 4.76 ms           | 1.32 ms | 41.61 ms | Pass (one outlier on first request — template compilation) |

## Step 4: Observations & Analysis

- All endpoints responded well within the 2-second target under demo/mock mode; the entire measured overhead is FastAPI routing + Pydantic validation + JSON-file logging, not AI inference.
- /api/quiz was the slowest of the AI endpoints (8.90 ms avg) because it also logs one record per generated quiz question to quizzes.json.
- The single 41.61 ms outlier on the home page was the first request, consistent with one-time Jinja2 template compilation.
- Live-mode performance (with a real GEMINI\_API\_KEY) was not measured in this test run and will be dominated by Gemini API network latency plus the built-in retry/backoff logic (up to 3 attempts); this is recommended as a follow-up test using a tool such as Locust or k6 once a production API key and hosting environment are available.
- The JSON flat-file data store performed adequately for this single-user, sequential test; concurrent-write behavior under multiple simultaneous users has not yet been load-tested and is flagged as a scalability risk (see Section 20, Scalability & Future Plan).